Subhash Suthar

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Profile

Final-year Electronics and Communication Engineering student with a strong foundation in Python and Digital Design, and experience in academic and software-based projects. Highly motivated to apply engineering principles to real-world applications while continuously developing technical and professional skills.

Education

Bachelor of Technology in Electronics and communication Engineering Aug 2022 – May 2026
Rajasthan Technical University (CGPA: 6.6) *Kota, India*

Technical Skills

Programming Languages: C, Python, C++(basic), HTML, CSS, Verilog
Technologies and Tools: Git, GitHub, MySQL, VS Code, Vivado
Soft Skills: Public Speaking, Communication, Problem Solving, Leadership, Critical Thinking

Internship

Analyser Instrument Co. Pvt. Ltd. June 24 – July 24

- Observed working of gas and water analyzer systems used for industrial monitoring.
- Assisted in basic testing and calibration of measuring instruments.
- Learned industrial safety practices and maintenance procedures.

Kota Super Thermal Power Plant May 25 – July 25

- Understood the working of thermal power generation and control system.
- Observed and studies various measuring and protection instruments used in plant.
- Learned basics of power plant automation and instrumentation.

Projects

Sudokraft – Sudoku Puzzle Generator and Solver May 2025

- Developed a **Python-based application** to generate and solve Sudoku puzzles using recursive backtracking
- Implemented constraint validation for rows, columns, and 3×3 subgrids using an efficient `is_valid()` function
- Used randomized 3×3 box-filling techniques to generate uniquely solvable puzzles with varying difficulty levels
- Applied object-oriented principles to design a modular **Sudoku solver class**
- Leveraged Python lambda functions for reusable subgrid-fill operations
- Enhanced recursive pruning techniques to improve backtracking efficiency

Serial Protocol Verification (UART, SPI, I2C) *March 2025*

- Built automated SystemVerilog self-checking testbenches to verify protocol timing and data integrity.
- Verified UART, SPI, and I2C protocols using assertions, scoreboards, and error injection techniques.

Extra-Curricular Activities

Mar 2024

Secured Runner-up position in Robo-Rush at JECRC (Mar 2024) by designing and controlling a robot for obstacle navigation and task-based challenges.